SMART MANUFACTURING AI

Visual Quality Inspection & Automated Defect Intelligence Platform

RESULT: FAIL

FAILED QUALITY AUDIT

Inspection ID:	INSP-1793369898-8E34B1	**Timestamp:**	2026-10-30 19:48:18
Component Name:	Tractor engine housing	**Similarity Score (SSIM):**	99.60%
Total Defect Regions:	1	**Affected Surface Area:**	0.409%
High-Severity Defects:	1	**Processing Duration:**	235.3 ms

Visual Evidence & Comparative Computer Vision Analysis

Figure A: Reference (Good Component)

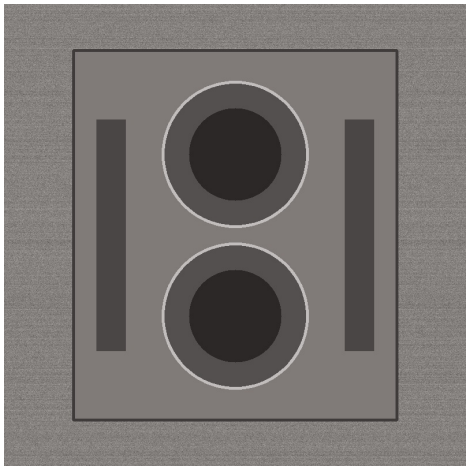


Figure B: Inspected Component

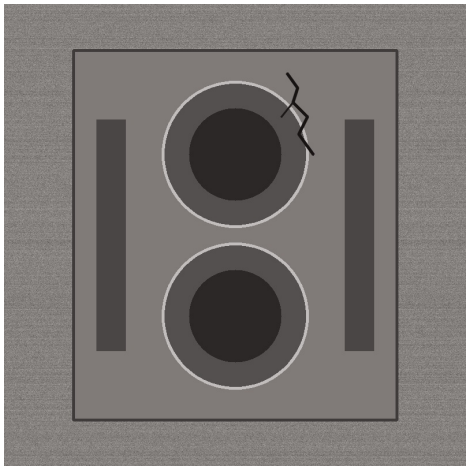


Figure C: SSIM Difference Heatmap

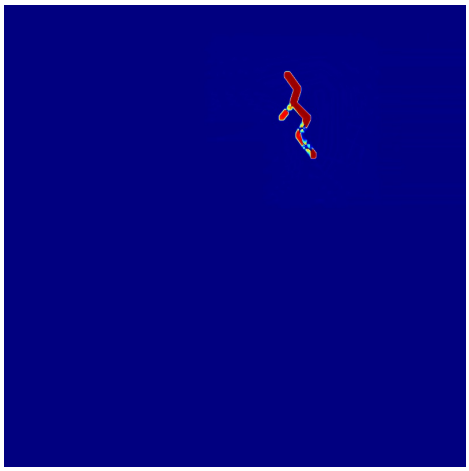
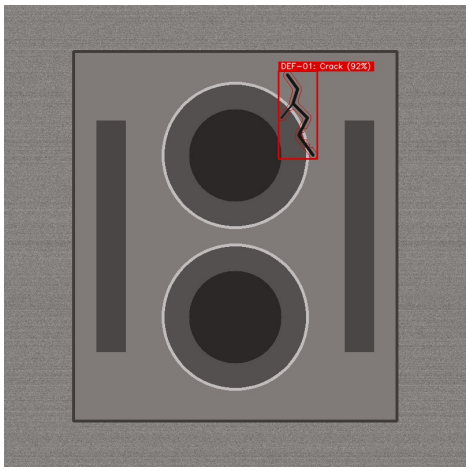


Figure D: Defect Highlighted & Annotated



Detected Defects & Anomaly Registry

Defect ID	Category	Severity	Score	Location (X, Y, W, H)	Area (% / px)	Description

DEF-01	**Crack**	**CRITICAL**	0.92	(475, 114) 67x152	0.41% (2618px)	Structural micro-fracture or crack with irregular propagation pattern.

AUTOMATED QUALITY ENGINEERING RECOMMENDATIONS:

- CRITICAL DEFECT: Structural integrity compromised. Quarantine component immediately to prevent field failure.
- Conduct non-destructive ultrasonic or magnetic particle inspection on adjacent lot items.
- Verify stamping/casting die pressures and cooling cycles on the manufacturing line.

Generated by Smart Manufacturing AI – Computer Vision Quality Inspection Intelligence Suite. Certified for plant floor auditor sign-off.